

Deep Learning for Skin Disease Detection Across Skin Tones: A Systematic Review

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Abstract—Skin diseases lead to a substantial global health burden, yet diagnostic outcomes remain unequal with patients of darker skin more frequently diagnosed at advanced stages. Deep learning (DL) has emerged as a promising approach for automated skin disease detection through dermoscopic and clinical image analysis. However, most of the models still underperform on darker skin tones as a result of imbalanced training data. This systematic literature review (SLR) aims to analyze and synthesize current research on DL-based skin disease detection across the Fitzpatrick skin type (FST) spectrum, with a focus on inclusivity and algorithmic fairness. Following the Kitchenham protocol and PRISMA guidelines, a total of 20 studies published between 2021-2026 were reviewed. This review finds a clear architectural shift from Convolutional Neural Networks (CNNs) toward hybrid CNN-Vision Transformer (ViT) and generative frameworks such as StyleGAN3 and CycleGAN, which are increasingly used to synthesize underrepresented dark skin samples. Most studies reported high overall accuracy; however, a synthesis of the results reveals a persistent performance gap ranging from 2% to 46% between lighter skin tones (FST I-III) and darker skin tones (FST IV-VI), depending on the specific diagnostic task and model architecture. With the underrepresentation of FST V-VI in public datasets, the subjectivity of Fitzpatrick labeling, and the limited validation on real-world clinical data, the challenges still remain. This review delivers insights into current progress, highlights existing gaps, and offers directions for future research toward developing equitable and trustworthy DL-based dermatology systems.

Keywords— deep learning, skin disease detection, dermatology, skin tone, Fitzpatrick scale, algorithmic fairness, systematic literature review

I. INTRODUCTION

Skin diseases represent a significant global health burden with a steadily increasing prevalence, reaching approximately 4.7 billion incident cases worldwide [1]. While the incidence of skin cancer is statistically lower in populations with darker skin tones compared to those with

lighter skin, patients with darker skin often face a disproportionately higher mortality rate. For instance, the five-year survival rate for melanoma among Black patients is approximately 70%, compared to 92% for White patients—a disparity primarily attributed to diagnosis at much more advanced stages [2]. Beyond malignancy, skin conditions encompass a wide range of inflammatory and infectious disorders, such as eczema, psoriasis, and scabies, which remain major public health challenges in regions with limited access to dermatologists, such as Southeast Asia and Africa [3]. The visual complexity of these conditions, which often manifest differently across various skin tones, makes accurate early diagnosis essential yet challenging even for trained clinicians.

Deep Learning has emerged as a pivotal tool to address these limitations through clinical image analysis and dermoscopy. Artificial neural networks are currently capable of achieving levels of accuracy comparable to expert dermatologists in recognizing various types of skin lesions [4]. Technically, the application of deep learning in this domain focuses on two primary tasks: classification to determine disease labels and detection to localize lesion areas on the skin surface [5]. Importantly, these automated systems are expected to be robust not only in identifying skin cancers but also across the entire spectrum of dermatological conditions to serve as effective self-assessment tools for the wider public.

Researchers have utilized diverse deep learning architectures to detect and classify this broad range of skin diseases. Early studies predominantly relied on Convolutional Neural Networks (CNNs), which demonstrated their superior capability in extracting spatial patterns from medical images but still struggled as network depth increased due to vanishing gradient problems [6]. This limitation urged a shift toward pre-trained architectures such as ResNet, whose skip connections enabled the training of deeper networks through residual connections and set a new performance benchmark in the field [7]. Building on this momentum, EfficientNet introduced a more principled scaling strategy—collectively optimizing network depth, width, and resolution. It became rapidly prevalent in dermatology research due to the superior accuracy parameter ratio [8]. More recently, Vision Transformers (ViT) have

introduced a different approach by capturing global relationships across an entire image rather than relying solely on local convolutional filters, which is particularly useful for conditions that do not follow consistent spatial patterns [9]. Across all these architectural generations, transfer learning has served as a consistent enabling strategy, allowing models to adapt from large-scale datasets to smaller, specialized dermatology datasets, to provide a practical solution when annotated medical data is limited [10].

However, behind these technical advances lies a critical issue of algorithmic bias that impacts the fairness of model performance across skin tone variations. The majority of existing models are trained on datasets dominated by light skin types (Fitzpatrick I–III), leading to significant performance disparities when applied to medium and dark skin tones (Fitzpatrick IV–VI). This bias extends beyond cancer detection to affect the diagnosis of inflammatory and infectious diseases, which exhibit distinct visual characteristics on darker skin tones. Without addressing this gap, AI-based dermatology tools risk amplifying healthcare disparities in regions like Indonesia and Southeast Asia, where darker skin tones predominate but remain underrepresented in current AI research.

Several systematic literature reviews for this topic have been published in recent years. However, their scope remains insufficient in addressing the specific dimension of skin tone inclusivity. Previous reviews have largely still focused on benchmarking model accuracy or comparing architectural performance in general terms [5], without systematically examining how model performance degrades across the Fitzpatrick skin type spectrum. Additionally, existing reviews seldom assess the demographic representativeness of the datasets used, nor do they explicitly identify the gaps relevant to underrepresented populations such as those with darker or medium skin tones.

Therefore, this study conducts a Systematic Literature Review following the Kitchenham protocol [11] to objectively evaluate the current state of deep learning technology in dermatology with a specific focus on skin tone inclusivity. Guided by the PICOC framework, this review answers four primary research questions:

- RQ1** What deep learning architectures and methods have been used for automated skin disease detection and classification across different skin tones?
- RQ2** What datasets are utilized in existing studies, and how representative are they in terms of skin tone distribution?
- RQ3** How does model performance vary across different Fitzpatrick skin types?
- RQ4** What are the key limitations and challenges reported in existing deep learning approaches for skin disease detection across diverse skin tones?

Thus, the ultimate goal of this study is to identify effective technical strategies for developing inclusive dermatology AI that is robust for all patients, regardless of their skin tone.

II. METHODOLOGY

A. Review Protocol and Registration

This systematic literature review was conducted in accordance with the guidelines proposed by Kitchenham [11], with the study selection process reported in conformity with the PRISMA standard. The review protocol was defined preceding the literature search process to ensure transparency, reproducibility, and methodological thoroughness.

B. Defining PICOC and Synonyms

The scope of the review was defined systematically with the PICOC (Population, Intervention, Comparison, Outcome, and Context) framework, employed to structure the research questions and guide the search strategy. The four research questions previously stated in the Introduction were inferred from this PICOC framework, which guided the search string construction as well. The PICOC elements and their corresponding synonyms were used to construct the search string, presented in Table I.

TABLE I. DEFINING PICOC AND SYNONYMS

PICOC Elements	Definition	Synonyms
Population	Refers to the patients with skin diseases across varying skin tones	Skin lesion, dermatological condition, skin cancer, eczema, psoriasis
Intervention	The primary methodology with deep learning models for automated detection and classification	CNN, ResNet, EfficientNet, ViT, neural network, image classification
Comparison	The model performance across different skin tones	Fitzpatrick scale, skin tone variation, dark skin, light skin
Outcome	The result of detection accuracy and fairness across skin types	Accuracy, AUC, F1-score, sensitivity, algorithmic bias
Context	The specific clinical image and dermoscopy analysis studies	Dermoscopy, clinical photography, medical imaging

C. Eligibility Criteria

The relevance and methodological quality of the included studies was ensured with a set of the eligibility criteria and were applied systematically during the full-text review stage. Eligible studies were peer-reviewed journal articles published in English between 2021 and April 2026, indexed in ScienceDirect, IEEE Xplore, PubMed, ACM Digital Library, or SpringerLink. Studies that qualified were required to use dermoscopic or clinical photographic image of human skin diseases as input data, employ deep learning as the primary method, such as CNN, ResNet, EfficientNet, and ViT. Studies were also required to report quantitative results with clear evaluation metrics and to explicitly mention skin tone, skin color, or Fitzpatrick skin type in any capacity.

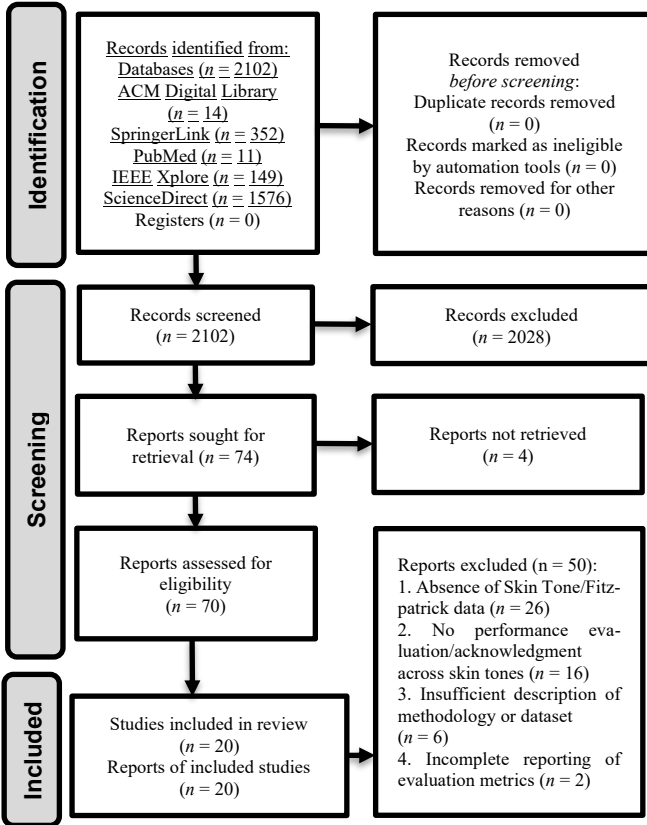
D. Information Sources and Search Strategy

A systematic search was conducted across IEEE Xplore, ScienceDirect, SpringerLink, ACM, and PubMed for studies published from January 2021 to April 2026. Furthermore, reference lists of the selected literature were manually screened to identify additional eligible studies.

The search strings were constructed based on three core concept clusters derived from the PICOC framework: (1) the population — skin disease and dermatological conditions; (2) the intervention — deep learning methods and architectures; and (3) the context — skin tone diversity and Fitzpatrick scale representation.

The search strings were adapted to the variations and synonyms such as: ("skin disease" OR "skin lesion" OR "dermatology" OR "skin condition") AND ("deep learning" OR "convolutional neural network" OR "CNN" OR "EfficientNet" OR "Vision Transformer" OR "ViT" OR "ResNet") AND ("skin tone" OR "skin color" OR "Fitzpatrick" OR "skin pigmentation" OR "dark skin" OR "skin of color"). Boolean operators (AND, OR) were used to combine terms properly. The search strategy was adapted for the syntax requirements of each database while preserving the same abstract coverage across three clusters. The complete study selection process is illustrated in Fig. 1.

FIG. 1. PRISMA FLOW DIAGRAM



E. Data Extraction

Data extraction was developed systematically and pilot tested on a subset of included studies that passed the criteria and quality assessment phases. Authors independently extracted data from the included study to ensure consistency and completeness based on the data extraction form. The extracted information covered general information (author, title, year, DOI) and four thematic categories aligned with the research questions.

F. Quality Assessment

A quality assessment was conducted to evaluate the potential risk of bias of the included study that passed the eligibility criteria. Each study was assessed using six custom

questions, scored on a scale of 1 to 3, where 1 indicated the criterion is not met, 2 indicates it is partially met, and 3 indicates it is fully met. The maximum achievable score is 18 and studies with a total score under 12.6 (70% of the maximum score) were excluded on the grounds of insufficient methodological quality.

QA3 and QA5 are designated as critical criteria due to the direct positioning with the core focus of this review. Studies that received a 1 score on both QA3 and QA5 were flagged for additional review. The six quality assessment criteria are as follows:

- QA1** Is the deep learning method clearly described and reproducible?
- QA2** Is the dataset used clearly described in terms of size, source, and composition?
- QA3** Does the study report skin tone or Fitzpatrick Skin Type (FST) information?
- QA4** Are evaluation metrics clearly reported (e.g., accuracy, AUC, sensitivity, F1-score)?
- QA5** Does the study evaluate or acknowledge performance differences across skin tones?
- QA6** Does the study identify future research directions or gaps related to model generalizability or skin tone diversity?

III. RESULTS AND DISCUSSION

This section synthesizes findings from 20 peer-reviewed studies to address the four research questions. The analysis reveals architectural trends, dataset limitations, performance disparities, and persistent technical challenges in developing equitable deep learning systems for skin disease detection across diverse skin tones.

RQ1: What deep learning architectures and methods have been used for automated skin disease detection and classification across different skin tones?

The reviewed studies demonstrate a clear evolution from traditional Convolutional Neural Networks toward more sophisticated hybrid and fairness-aware architectures. Among the 20 studies, 25% ($n = 5$) employed hybrid CNN-Vision Transformer (ViT) architectures, reflecting a shift toward models capable of capturing both local texture features and global contextual patterns. Generative frameworks, particularly StyleGAN3 and CycleGAN, emerged in 25% ($n = 5$) of studies as a dominant strategy for synthetic data augmentation targeting underrepresented darker skin tones. Transfer learning remains ubiquitous, with 90% ($n = 18$) of studies leveraging pre-trained weights from ImageNet, though recent work has begun exploring self-supervised learning (SSL) and masked image modeling (MIM) as fairness-aware alternatives to mitigate inherited bias from pre-training datasets.

Bias mitigation techniques can be categorized into four primary strategies. First, *generative augmentation* utilizes GANs or diffusion models to synthesize dark skin samples to balance the dataset. Second, *feature disentanglement* aims to isolate disease-related features from skin tone attributes through specialized loss functions. Third, *domain adaptation* facilitates the transfer of knowledge from majority to minority skin tone distributions. Finally, *fairness-aware*

training incorporates explicit constraints, such as Equal Opportunity, Demographic Parity, or adversarial debiasing during model optimization. Notably, the integration of explainability mechanisms (Grad-CAM, attention maps) appeared in 40% ($n = 8$) of studies, signaling growing awareness of the need for interpretable AI in clinical settings.

The architectural shift toward hybrid models reflects the inherent challenge of skin disease detection on darker skin tones, where lesions often exhibit lower contrast against surrounding tissue. Vision Transformers' ability to model long-range dependencies proves particularly valuable for conditions where diagnostic features are distributed non-locally across the image. However, the computational cost of ViT-based models remains a barrier for real-time clinical deployment, driving continued interest in lightweight hybrid architectures such as MobileViT that balance accuracy with efficiency.

TABLE II. SUMMARY OF DEEP LEARNING ARCHITECTURES GROUPED BY CATEGORY

Architecture Category	Representative Papers	Key Characteristics
Traditional CNN	VGG-16 [13, 20], ResNet [12, 15, 18, 21, 28, 31]	Baseline architectures; most combined with transfer learning from ImageNet
Hybrid CNN-ViT	DermNet [17], Dual-Stream CNN-ViT [14], CAS-ViT [29], MobileViT [24, 26]	Capture both local and global features; particularly effective for non-uniform lesion patterns
Generative Models	StyleGAN3 [13, 26], CycleGAN [25], Diffusion [12], VAE [18]	Synthetic augmentation of underrepresented FST V-VI; address class imbalance
Ensemble Methods	SENet + SE-ResNeXt + ResNeSt [31]	Multi-model voting to improve robustness across diverse inputs
Lightweight Architectures	TiSkID-Net [23]	Optimized for mobile deployment with minimal parameters
Fairness-Aware Architectures	AttEN [22], DB-VAE [18], SITL + FEDL [29]	Explicit fairness constraints through loss functions or feature disentanglement

RQ2: What datasets are used in existing studies, and how representative are they in terms of skin tone distribution?

The analysis reveals a concentration of research around a small number of publicly available datasets, with HAM10000, ISIC-2019, and Fitzpatrick17k collectively appearing in 65% ($n = 13$) of reviewed studies. A critical finding is the severe underrepresentation of darker skin tones: among studies that reported Fitzpatrick Skin Type (FST) distribution, FST V-VI comprised only 8–12% of training data on average, with some datasets containing less than 1% dark skin representation. The analysis reveals that while only 30% ($n = 6$) of the reviewed studies utilized datasets with pre-existing expert-provided FST labels, the remaining studies ($n = 14$) relied on automated estimation methods or proxy measures to infer skin tone information for their evaluations.

Geographic concentration further compounds the representativeness problem. Approximately 80% ($n = 16$) of datasets originated from North America or Europe, with only

one study [31] utilizing data exclusively from Asia (South Korea). No studies incorporated datasets from Southeast Asia, Sub-Saharan Africa, or South Asia, despite these regions harboring the majority of FST IV-VI populations globally. Class imbalance extends beyond skin tone to disease distribution, with common benign lesions (melanocytic nevi) vastly outnumbering rare but clinically critical conditions such as acral lentiginous melanoma, which disproportionately affects darker-skinned populations.

TABLE III. DATASETS GROUPED BY PRIMARY SOURCE AND SKIN TONE COVERAGE

Dataset Name	Studies Using This Dataset	Reported FST Coverage	Known Limitations
Fitzpatrick17k	[13, 15, 17, 18, 22, 24]	FST I-VI labeled; predominantly FST I-III (~70%); FST V-VI <10%	Subjective labeling, limited to 114 conditions
HAM10000	[14, 26, 30]	FST not labeled in original; estimated 90%+ light skin	No FST metadata; European population; class imbalance (67% nevi)
ISIC-2019 / ISIC-2020	[15, 22, 24, 25, 29, 30]	FST not labeled; estimated 85%+ FST I-III	Dominated by melanoma; limited diversity in inflammatory conditions
DDI (Diverse Dermatology Images)	[17, 26]	Explicitly diverse; ~40% FST IV-VI	Small size (~650 images); limited disease classes
PAD-UFES-20	[25, 29]	Brazilian population; 20% darker skin	Only 2,298 images; binary classification (benign/malignant)
SkinCon (Fitzpatrick17k + DDI merged)	[17]	Balanced: White 1340, Brown 1154, Black 1149	Curated subset; not publicly available
Custom / Institutional	[19, 20, 21, 23, 27, 28, 31]	Varies widely; [31] exclusively FST III-IV (Asian); others predominantly light	Limited external validation; ethical/privacy restrictions on sharing

Addressing pervasive dataset limitations, 70% ($n = 14$) of the selected studies utilized at least one imbalance mitigation technique. Synthetic augmentation via GANs or style transfer emerged as the primary strategy ($n = 8$), followed by oversampling methods such as SMOTE or random oversampling in six studies. Furthermore, five researchers implemented stratified sampling protocols to ensure balanced representation across FST groups, while three studies focused on domain adaptation to transfer knowledge from majority to minority distributions. However, the reliance on synthetic data remains controversial due to potential compromises in morphological fidelity, where generated images may alter lesion textures or boundary characteristics in ways that fail to reflect authentic disease presentation on darker skin.

RQ3: How does model performance vary across different Fitzpatrick skin types?

Quantitative analysis of reported performance metrics reveals a persistent accuracy gap between lighter (FST I-III) and darker (FST IV-VI) skin tones across the majority of reviewed studies. Among the 18 studies that performed stratification (including those using both explicit labels and estimated skin-tone data), 72% ($n = 13$) documented lower accuracy, sensitivity, or segmentation quality on darker skin tones, with performance degradation ranging from 2% to 46% depending on the task and model architecture. Segmentation tasks exhibited particularly severe disparities, with Dice Score Coefficient (DSC) dropping from 0.90 on light skin to 0.74 on dark skin in one study [16] — a 16-point decline that would significantly impact clinical utility for lesion boundary delineation.

Encouragingly, studies implementing explicit bias mitigation techniques demonstrated substantial improvements. For instance, fairness-aware approaches reduced performance gaps to as low as 2% [25], and one lightweight model [23] achieved near-parity (<2% gap) across all skin tone groups. However, these successes often came with trade-offs: fairness-constrained models sometimes exhibited slightly lower overall accuracy compared to unconstrained baselines, and computational complexity increased in methods employing adversarial debiasing or multi-objective loss functions. A notable outlier finding from [27] suggested that condition distribution shifts — rather than skin tone per se — may be the primary driver of generalization failure in real-world deployment, though this hypothesis requires validation across broader geographic and demographic contexts.

TABLE IV. PERFORMANCE SUMMARY BY EVALUATION METRIC AND SKIN TONE

Metric	Best Performance (FST I-III)	Best Performance (FST IV-VI)	Models Achieving <5% Gap
Accuracy	98.55% [30] — light skin only	97% [23] — minimal gap	[23] TiSkiD-Net (gap <2%), [25] CycleGAN adaptation (gap 2%)
F1-Score	0.98 [30] — light skin only	0.96 [23] — near-parity	[22] AttEN (improved fairness metrics), [29] CAS-ViT-SIRFDL
AUC/ROC	0.91 [26] — after StyleGAN3 augmentation	0.86 [24] — federated learning with IRM	[26] MobileViT + StyleGAN3 (>10% fairness improvement)
Dice Similarity Coefficient	0.90 [16]	0.74 [16]	[17] DermNet (minimal gap via lesion-only segmentation)
Fairness Metrics (EOD, DI)	N/A	Baseline EOD 0.21 → 0.09 [24]	[24] Federated Learning + IRM (50% fairness improvement), [22] AttEN

The observed disparities stem from multiple compounding factors. First, dataset imbalance leads models to optimize primarily for the majority class (light skin), with insufficient exposure to darker skin patterns during training. Second, low contrast between lesions and surrounding tissue on darker skin reduces the signal available for feature

extraction, particularly for CNN-based models relying on edge detection. Third, annotation quality may be systematically lower for darker skin images if ground-truth labels were generated by clinicians with less experience diagnosing on diverse skin tones — a form of label bias that propagates into model learning.

Interestingly, the magnitude of performance gap varies by disease type. Studies focusing on melanoma detection showed larger disparities (10–20% accuracy gap) compared to those targeting inflammatory conditions like eczema or psoriasis (5–10% gap), possibly because inflammatory markers (erythema, scaling) manifest differently on darker skin and require domain-specific feature representations that current architectures inadequately capture.

RQ4: What are the key limitations and challenges reported in existing deep learning approaches for skin disease detection across diverse skin tones?

The reviewed literature reveals four major categories of challenges that collectively impede the development of equitable skin disease detection systems: (1) data quality and representativeness limitations, (2) model architecture and generalization constraints, (3) evaluation methodology gaps, and (4) clinical translation barriers.

a. Data Quality and Representativeness Limitations

The most frequently cited challenge (85%, $n = 17$) is the underrepresentation of darker skin tones in public datasets, which drives reliance on synthetic generation that may fail to reproduce authentic disease morphology. FST labels also remain subjective, relying on clinician judgment or self-report, while objective alternatives such as ITA require standardized lighting conditions rarely available in clinical practice. In addition, 40% ($n = 8$) of studies reported incomplete or biased metadata.

b. Model Architecture and Generalization Constraints

Architectural limitations appeared in 55% ($n = 11$) of studies. Models pre-trained on ImageNet inherit light-skin bias that fine-tuning only partially mitigates without fairness-aware pre-training [15]. Segmentation tasks are particularly affected because standard color-space normalization (CIELAB) performs poorly on low-contrast lesion boundaries in darker skin [16], while lightweight mobile models face an accuracy-fairness trade-off where compression disproportionately degrades darker-skin performance [23].

c. Evaluation Methodology Gaps

A challenge identified in 60% ($n = 12$) of studies is the absence of dermatology-specific fairness evaluation protocols. Although general metrics such as Equal Opportunity and Demographic Parity are increasingly reported, their clinical relevance remains unclear. Moreover, only 25% ($n = 5$) of studies stratified performance by both skin tone and disease type, and 70% ($n = 14$) relied on retrospective datasets, with only one prospective real-world validation study [31], limiting generalizability.

d. Clinical Translation Barriers

Clinical translation, discussed in 45% ($n = 9$) of studies, remains constrained by regulatory hurdles, insufficient explainability, clinician over-reliance on erroneous AI suggestions [20], and privacy restrictions that limit access to diverse datasets. Although Grad-

CAM and attention maps are increasingly adopted, they often highlight image artifacts rather than clinically relevant features. Only one study reported pursuing FDA approval [26].

IV. CONCLUSIONS

In conclusion, this systematic literature review of 20 studies highlights both the progress and the remaining challenges in applying deep learning to skin disease detection across diverse skin tones. Current research demonstrates a shift from conventional CNNs toward hybrid CNN-Vision Transformer and generative approaches, yet a persistent performance gap on darker skin tones remains, largely due to the underrepresentation of minority skin types in public datasets. Future work should prioritize objective skin tone assessment, diverse real-world datasets, and transparent evaluation frameworks incorporating Explainable AI and patient metadata. Achieving algorithmic fairness in dermatology ultimately requires more inclusive dataset design to ensure equitable outcomes across all skin tones.

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